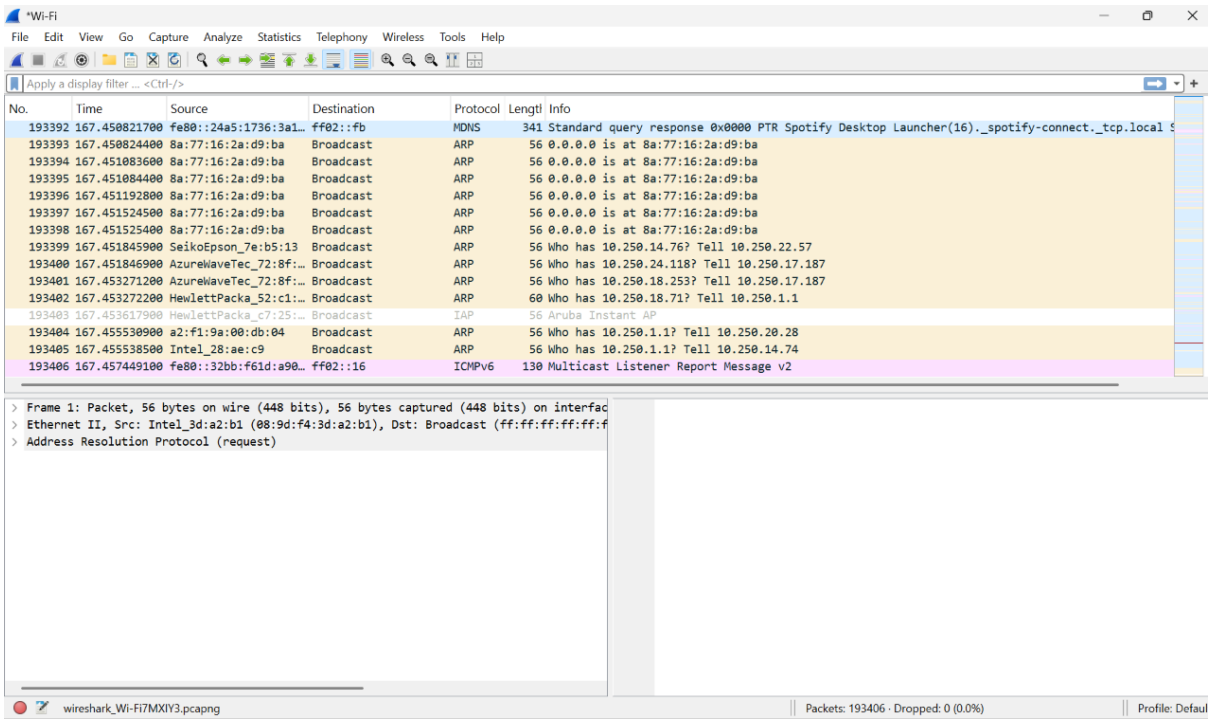
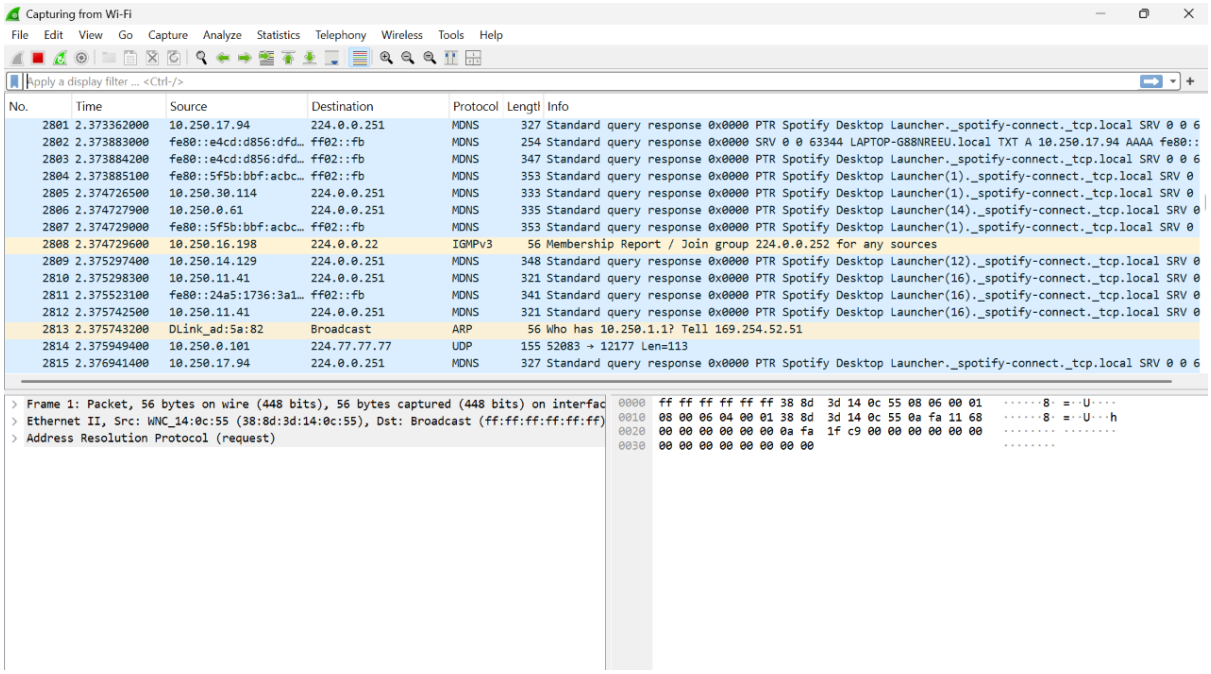
SKILL-1

STEP 1: Installation of wire shark

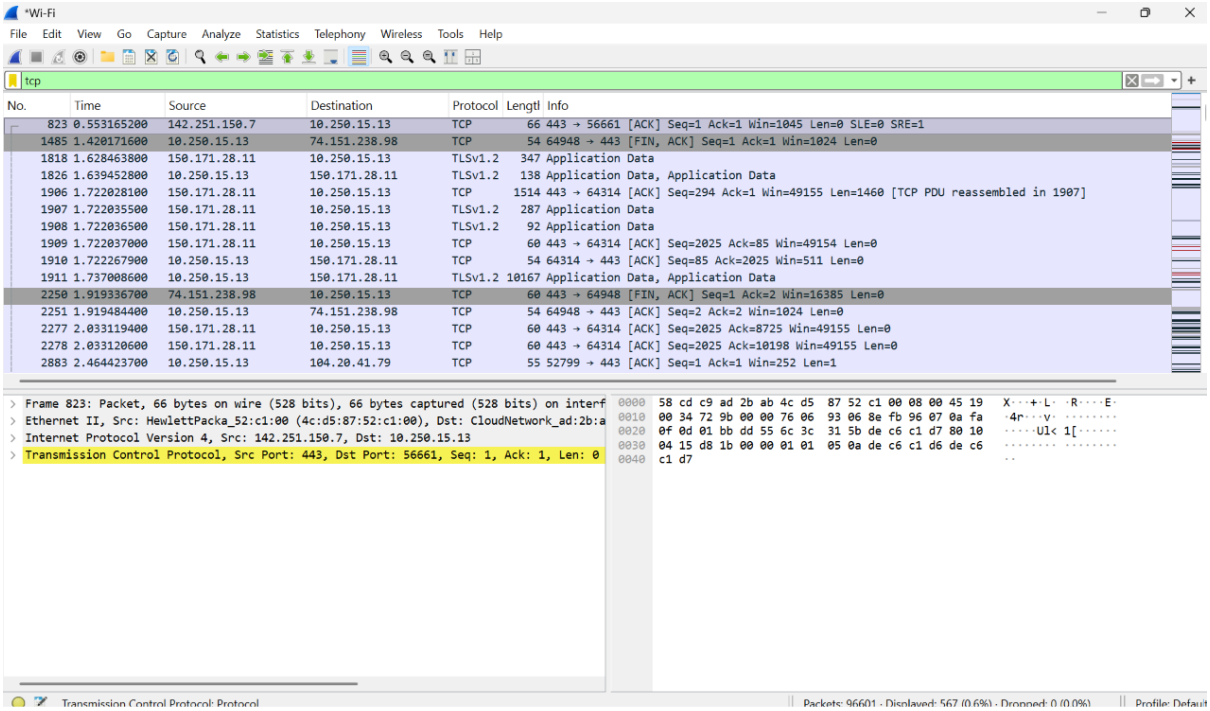
STEP 2:

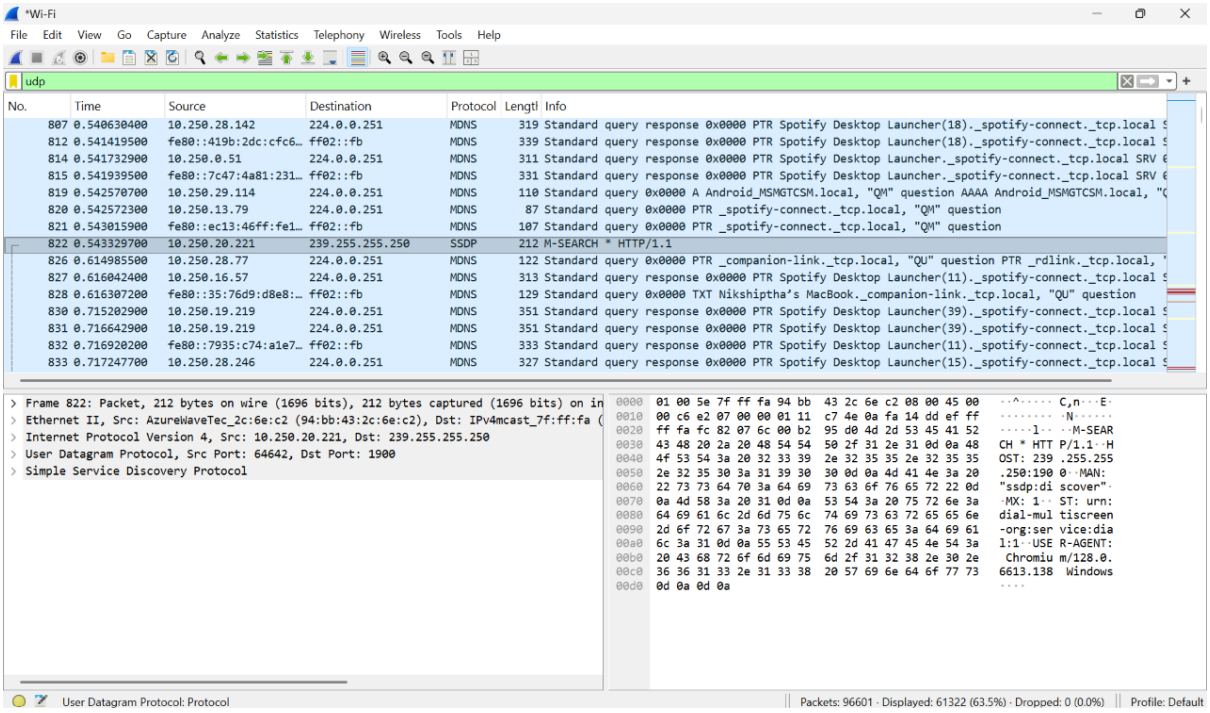


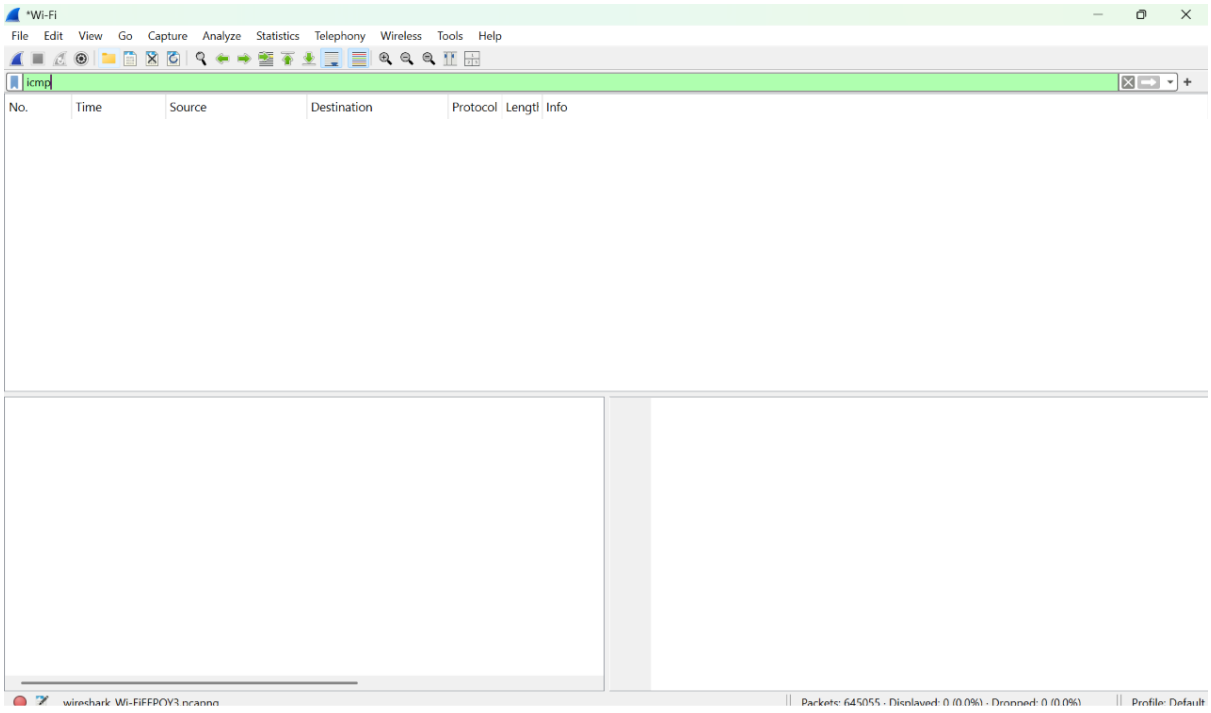
STEP 3:

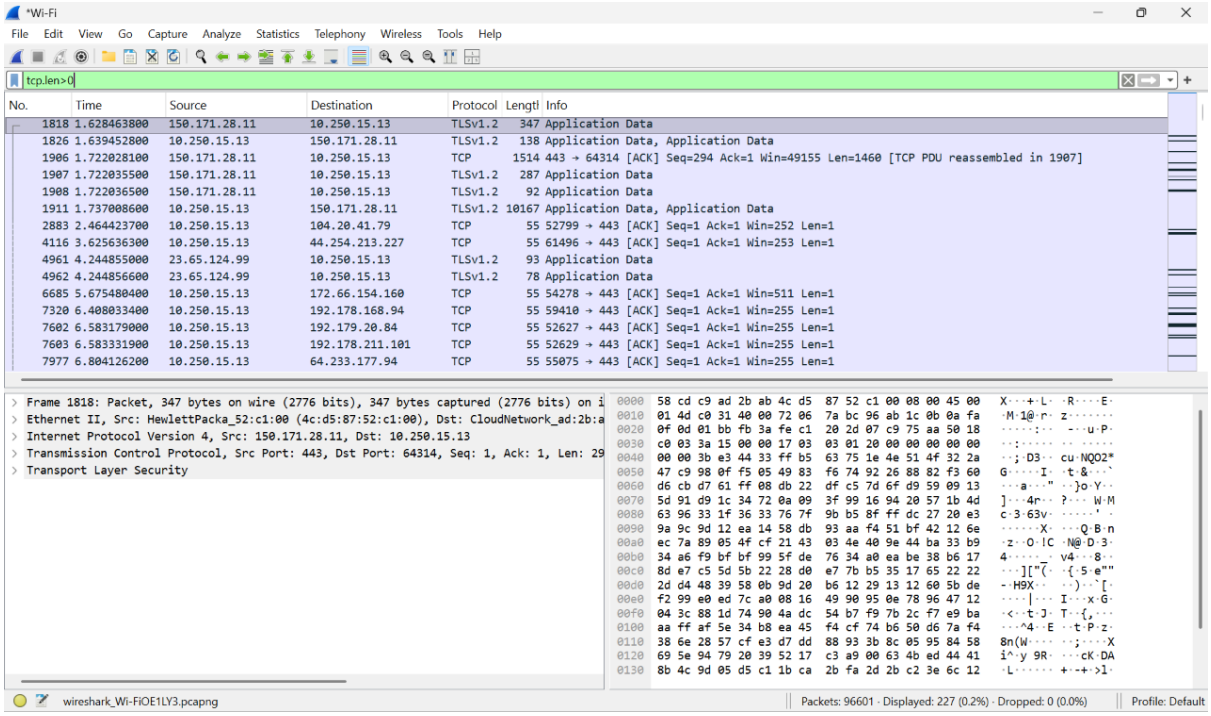


STEP 4 :

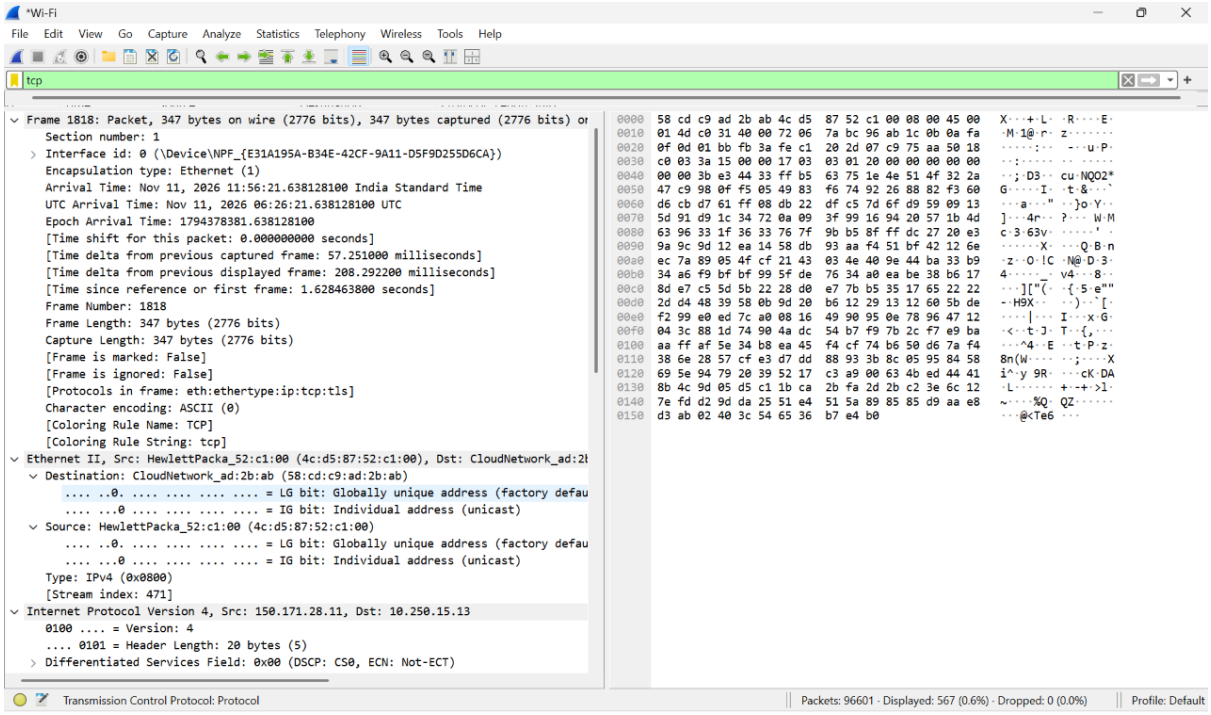

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
823 0.553165200 142.251.150.7 10.250.15.13 TCP 66 443 → 56661 [ACK] Seq=1 Ack=1 Win=1045 Len=0 SLE=0 SRE=1
1485 1.420171600 10.250.15.13 74.151.238.98 TCP 54 64948 → 443 [FIN, ACK] Seq=1 Ack=1 Win=1024 Len=0
1818 1.628463800 150.171.28.11 10.250.15.13 TLSv1.2 347 Application Data
1826 1.639452800 10.250.15.13 150.171.28.11 TLSv1.2 138 Application Data, Application Data
1906 1.722028100 150.171.28.11 10.250.15.13 TCP 1514 443 → 64314 [ACK] Seq=294 Ack=1 Win=49155 Len=1460 [TCP PDU reassembled in 1907]
1907 1.722035500 150.171.28.11 10.250.15.13 TLSv1.2 287 Application Data
1908 1.722036500 150.171.28.11 10.250.15.13 TLSv1.2 92 Application Data
1909 1.722037000 150.171.28.11 10.250.15.13 TCP 60 443 → 64314 [ACK] Seq=2025 Ack=85 Win=49154 Len=0
1910 1.722267900 10.250.15.13 150.171.28.11 TCP 54 64314 → 443 [ACK] Seq=85 Ack=2025 Win=511 Len=0
1911 1.737008600 10.250.15.13 150.171.28.11 TLSv1.2 10167 Application Data, Application Data
2250 1.919336700 74.151.238.98 10.250.15.13 TCP 60 443 → 64948 [FIN, ACK] Seq=1 Ack=2 Win=16385 Len=0
2251 1.919484400 10.250.15.13 74.151.238.98 TCP 54 64948 → 443 [ACK] Seq=2 Ack=2 Win=1024 Len=0
2277 2.033119400 150.171.28.11 10.250.15.13 TCP 60 443 → 64314 [ACK] Seq=2025 Ack=8725 Win=49155 Len=0
2278 2.033120600 150.171.28.11 10.250.15.13 TCP 60 443 → 64314 [ACK] Seq=2025 Ack=10198 Win=49155 Len=0
2883 2.464423700 10.250.15.13 104.20.41.79 TCP 55 52799 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
Frame 823: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interf
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_ad:2b:a
Internet Protocol Version 4, Src: 142.251.150.7, Dst: 10.250.15.13
Transmission Control Protocol, Src Port: 443, Dst Port: 56661, Seq: 1, Ack: 1, Len: 0
0000 58 cd c9 ad 2b ab 4c d5 87 52 c1 00 08 00 45 19 X···+·L· ·R····E·
0010 00 34 72 9b 00 00 76 06 93 06 8e fb 96 07 0a fa ·4r···v· ····
0020 0f 0d 01 bb dd 55 6c 3c 31 5b de c6 c1 d7 80 10 ·····Ul< 1[····
0030 04 15 d8 1b 00 00 01 01 05 0a de c6 c1 d6 de c6 ····· ····
0040 c1 d7 ··
Transmission Control Protocol: Protocol
Packets: 96601 · Displayed: 567 (0.6%) · Dropped: 0 (0.0%)
Profile: Default

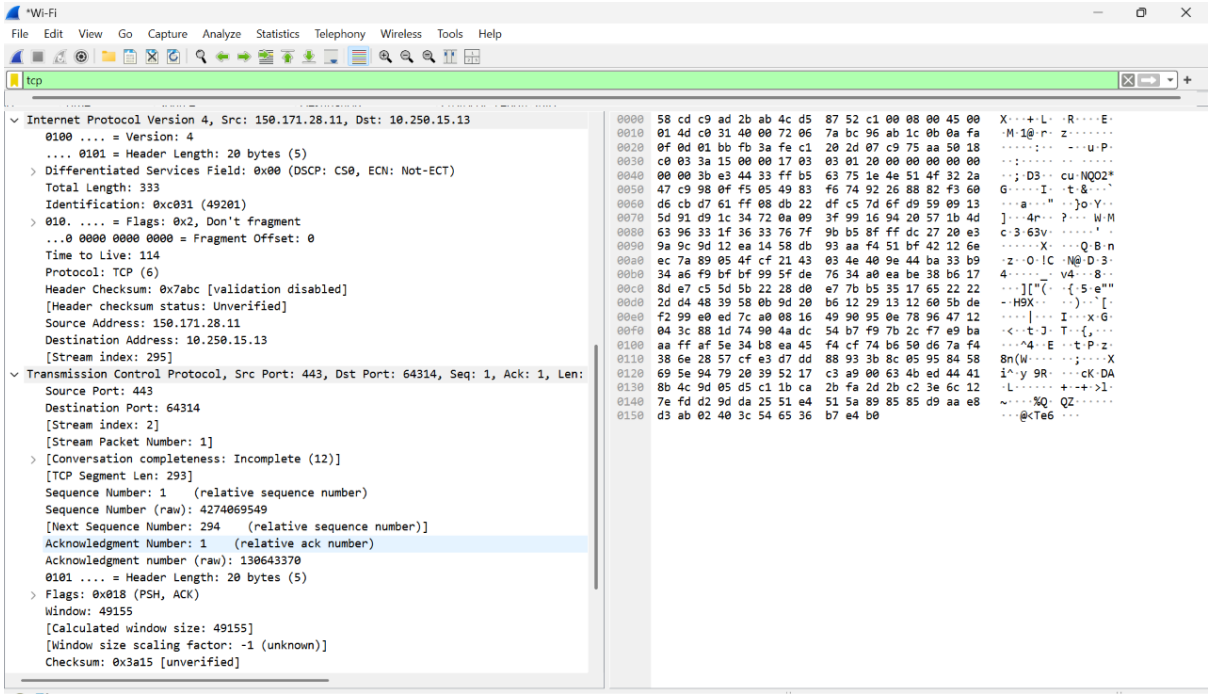

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Lengtl Info
807 0.540630400 10.250.28.142 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher(18)._spotify-connect._tcp.local
812 0.541419500 fe80::419b:2dc:cfc6… ff02::fb MDNS 339 Standard query response 0x0000 PTR Spotify Desktop Launcher(18)._spotify-connect._tcp.local
814 0.541732900 10.250.0.51 224.0.0.251 MDNS 311 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV
815 0.541939500 fe80::7c47:4a81:231… ff02::fb MDNS 331 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV
819 0.542570700 10.250.29.114 224.0.0.251 MDNS 110 Standard query 0x0000 A Android_MSMGTCSM.local, "QM" question AAAA Android_MSMGTCSM.local, "
820 0.542572300 10.250.13.79 224.0.0.251 MDNS 87 Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
821 0.543015900 fe80::ec13:46ff:fe1… ff02::fb MDNS 107 Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
822 0.543329700 10.250.20.221 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
826 0.614985500 10.250.28.77 224.0.0.251 MDNS 122 Standard query 0x0000 PTR _companion-link._tcp.local, "QU" question PTR _rdlink._tcp.local,
827 0.616042400 10.250.16.57 224.0.0.251 MDNS 313 Standard query response 0x0000 PTR Spotify Desktop Launcher(11)._spotify-connect._tcp.local
828 0.616307200 fe80::35:76d9:d8e8:… ff02::fb MDNS 129 Standard query 0x0000 TXT Nikshiptha's MacBook._companion-link._tcp.local, "QU" question
830 0.715202900 10.250.19.219 224.0.0.251 MDNS 351 Standard query response 0x0000 PTR Spotify Desktop Launcher(39)._spotify-connect._tcp.local
831 0.716642900 10.250.19.219 224.0.0.251 MDNS 351 Standard query response 0x0000 PTR Spotify Desktop Launcher(39)._spotify-connect._tcp.local
832 0.716920200 fe80::7935:c74:a1e7… ff02::fb MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(11)._spotify-connect._tcp.local
833 0.717247700 10.250.28.246 224.0.0.251 MDNS 327 Standard query response 0x0000 PTR Spotify Desktop Launcher(15)._spotify-connect._tcp.local
Frame 822: Packet, 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits) on in
Ethernet II, Src: AzureWaveTec_2c:6e:c2 (94:bb:43:2c:6e:c2), Dst: IPv4mcast_7f:ff:fa (
Internet Protocol Version 4, Src: 10.250.20.221, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 64642, Dst Port: 1900
Simple Service Discovery Protocol
0000 01 00 5e 7f ff fa 94 bb 43 2c 6e c2 08 00 45 00 ··^····· C,n···E·
0010 00 c6 e2 07 00 00 01 11 c7 4e 0a fa 14 dd ef ff ·····N·····
0020 ff fa fc 82 07 6c 00 b2 95 d0 4d 2d 53 45 41 52 ·····l·· ··M-SEAR
0030 43 48 20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48 CH * HTT P/1.1··H
0040 4f 53 54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35 OST: 239 .255.255
0050 2e 32 35 30 3a 31 39 30 30 0d 0a 4d 41 4e 3a 20 .250:190 0··MAN:
0060 22 73 73 64 70 3a 64 69 73 63 6f 76 65 72 22 0d "ssdp:di scover"·
0070 0a 4d 58 3a 20 31 0d 0a 53 54 3a 20 75 72 6e 3a ·MX: 1·· ST: urn:
0080 64 69 61 6c 2d 6d 75 6c 74 69 73 63 72 65 65 6e dial-mul tiscreen
0090 2d 6f 72 67 3a 73 65 72 76 69 63 65 3a 64 69 61 -org:ser vice:dia
00a0 6c 3a 31 0d 0a 55 53 45 52 2d 41 47 45 4e 54 3a l:1··USE R-AGENT:
00b0 20 43 68 72 6f 6d 69 75 6d 2f 31 32 38 2e 30 2e Chromiu m/128.0.
00c0 36 36 31 33 2e 31 33 38 20 57 69 6e 64 6f 77 73 6613.138 Windows
00d0 0d 0a 0d 0a ····
User Datagram Protocol: Protocol
Packets: 96601 · Displayed: 61322 (63.5%) · Dropped: 0 (0.0%)
Profile: Default

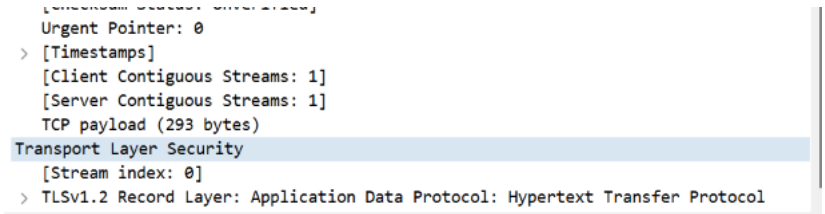

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Lengtl Info


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
No. Time Source Destination Protocol Lengtl Info
1818 1.628463800 150.171.28.11 10.250.15.13 TLSv1.2 347 Application Data
1826 1.639452800 10.250.15.13 150.171.28.11 TLSv1.2 138 Application Data, Application Data
1906 1.722028100 150.171.28.11 10.250.15.13 TCP 1514 443 → 64314 [ACK] Seq=294 Ack=1 Win=49155 Len=1460 [TCP PDU reassembled in 1907]
1907 1.722035500 150.171.28.11 10.250.15.13 TLSv1.2 287 Application Data
1908 1.722036500 150.171.28.11 10.250.15.13 TLSv1.2 92 Application Data
1911 1.737008600 10.250.15.13 150.171.28.11 TLSv1.2 10167 Application Data, Application Data
2883 2.464423700 10.250.15.13 104.20.41.79 TCP 55 52799 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
4116 3.625636300 10.250.15.13 44.254.213.227 TCP 55 61496 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
4961 4.244855000 23.65.124.99 10.250.15.13 TLSv1.2 93 Application Data
4962 4.244856600 23.65.124.99 10.250.15.13 TLSv1.2 78 Application Data
6685 5.675480400 10.250.15.13 172.66.154.160 TCP 55 54278 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1
7320 6.408033400 10.250.15.13 192.178.168.94 TCP 55 59410 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
7602 6.583179000 10.250.15.13 192.179.20.84 TCP 55 52627 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
7603 6.583331900 10.250.15.13 192.178.211.101 TCP 55 52629 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
7977 6.804126200 10.250.15.13 64.233.177.94 TCP 55 55075 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
Frame 1818: Packet, 347 bytes on wire (2776 bits), 347 bytes captured (2776 bits) on i
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_ad:2b:a
Internet Protocol Version 4, Src: 150.171.28.11, Dst: 10.250.15.13
Transmission Control Protocol, Src Port: 443, Dst Port: 64314, Seq: 1, Ack: 1, Len: 29
Transport Layer Security
wireshark_Wi-FiOE1LY3.pcapng
Packets: 96601 · Displayed: 227 (0.2%) · Dropped: 0 (0.0%)
Profile: Default

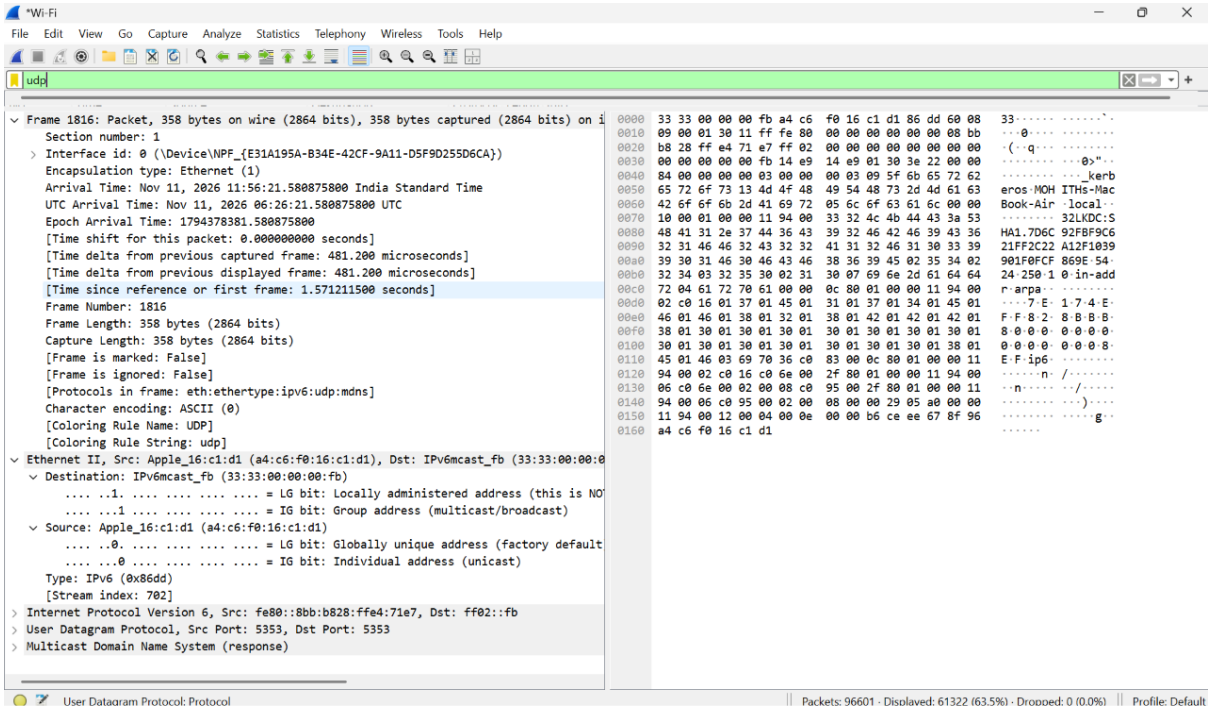
STEP 5: TCP

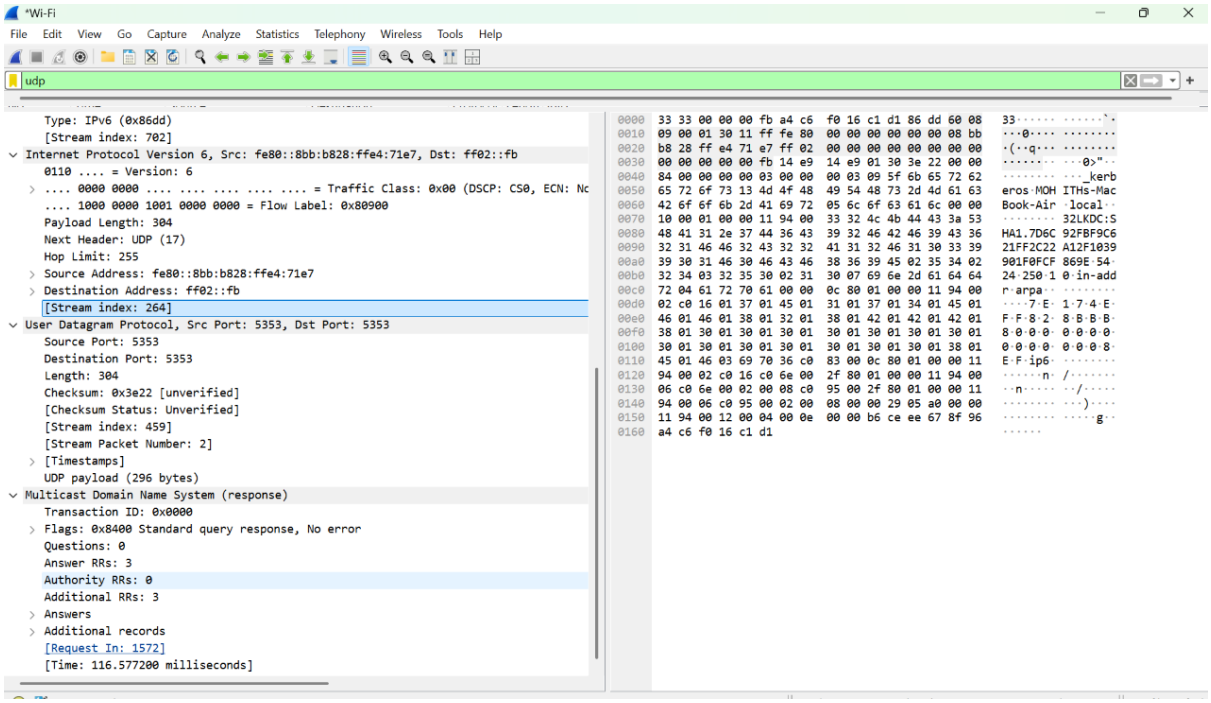

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
Frame 1818: Packet, 347 bytes on wire (2776 bits), 347 bytes captured (2776 bits) or
Section number: 1
Interface id: 0 (\Device\NPF_{E31A195A-B34E-42CF-9A11-D5F9D255D6CA})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 11:56:21.638128100 India Standard Time
UTC Arrival Time: Nov 11, 2026 06:26:21.638128100 UTC
Epoch Arrival Time: 1794378381.638128100
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 57.251000 milliseconds]
[Time delta from previous displayed frame: 208.292200 milliseconds]
[Time since reference or first frame: 1.628463800 seconds]
Frame Number: 1818
Frame Length: 347 bytes (2776 bits)
Capture Length: 347 bytes (2776 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_ad:2b
Destination: CloudNetwork_ad:2b:ab (58:cd:c9:ad:2b:ab)
.... ..0. .... = LG bit: Globally unique address (factory defau
....0 .... = IG bit: Individual address (unicast)
Source: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory defau
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 471]
Internet Protocol Version 4, Src: 150.171.28.11, Dst: 10.250.15.13
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 58 cd c9 ad 2b ab 4c d5 87 52 c1 00 08 00 45 00
0010 01 4d c0 31 40 00 72 06 7a bc 96 ab 1c 0b 0a fa
0020 0f 0d 01 bb fb 3a fe c1 20 2d 07 c9 75 aa 50 18
0030 c0 03 3a 15 00 00 17 03 03 01 20 00 00 00 00 00
0040 00 00 3b e3 44 33 ff b5 63 75 1e 4e 51 4f 32 2a
0050 47 c9 98 0f f5 05 49 83 f6 74 92 26 88 82 f3 60
0060 d6 cb d7 61 ff 08 db 22 df c5 7d 6f d9 59 09 13
0070 5d 91 d9 1c 34 72 0a 09 3f 99 16 94 20 57 1b 4d
0080 63 96 33 1f 36 33 76 7f 9b b5 8f ff dc 27 20 e3
0090 9a 9c 9d 12 ea 14 58 db 93 aa f4 51 bf 42 12 6e
00a0 ec 7a 89 05 4f cf 21 43 03 4e 40 9e 44 ba 33 b9
00b0 34 a6 f9 bf bf 99 5f de 76 34 a0 ea be 38 b6 17
00c0 8d e7 c5 5d 5b 22 28 d0 e7 7b b5 35 17 65 22 22
00d0 2d d4 48 39 58 0b 9d 20 b6 12 29 13 12 60 5b de
00e0 f2 99 e0 ed 7c a0 08 16 49 90 95 0e 78 96 47 12
00f0 04 3c 88 1d 74 90 4a dc 54 b7 f9 7b 2c f7 e9 ba
0100 aa ff af 5e 34 b8 ea 45 f4 cf 74 b6 50 d6 7a f4
0110 38 6e 28 57 cf e3 d7 dd 88 93 3b 8c 05 95 84 58
0120 69 5e 94 79 20 39 52 17 c3 a9 00 63 4b ed 44 41
0130 8b 4c 9d 05 d5 c1 1b ca 2b fa 2d 2b c2 3e 6c 12
0140 7e fd d2 9d da 25 51 e4 51 5a 89 85 85 d9 aa e8
0150 d3 ab 02 40 3c 54 65 36 b7 e4 b0
Transmission Control Protocol: Protocol
Packets: 96601 · Displayed: 567 (0.6%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
Internet Protocol Version 4, Src: 150.171.28.11, Dst: 10.250.15.13
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 333
Identification: 0xc031 (49201)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 114
Protocol: TCP (6)
Header Checksum: 0x7abc [validation disabled]
[Header checksum status: Unverified]
Source Address: 150.171.28.11
Destination Address: 10.250.15.13
[Stream index: 295]
Transmission Control Protocol, Src Port: 443, Dst Port: 64314, Seq: 1, Ack: 1, Len:
Source Port: 443
Destination Port: 64314
[Stream index: 2]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (12)]
[TCP Segment Len: 293]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4274069549
[Next Sequence Number: 294 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 130643370
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 49155
[Calculated window size: 49155]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x3a15 [unverified]
0000 58 cd c9 ad 2b ab 4c d5 87 52 c1 00 08 00 45 00
0010 01 4d c0 31 40 00 72 06 7a bc 96 ab 1c 0b 0a fa
0020 0f 0d 01 bb fb 3a fe c1 20 2d 07 c9 75 aa 50 18
0030 c0 03 3a 15 00 00 17 03 03 01 20 00 00 00 00 00
0040 00 00 3b e3 44 33 ff b5 63 75 1e 4e 51 4f 32 2a
0050 47 c9 98 0f f5 05 49 83 f6 74 92 26 88 82 f3 60
0060 d6 cb d7 61 ff 08 db 22 df c5 7d 6f d9 59 09 13
0070 5d 91 d9 1c 34 72 0a 09 3f 99 16 94 20 57 1b 4d
0080 63 96 33 1f 36 33 76 7f 9b b5 8f ff dc 27 20 e3
0090 9a 9c 9d 12 ea 14 58 db 93 aa f4 51 bf 42 12 6e
00a0 ec 7a 89 05 4f cf 21 43 03 4e 40 9e 44 ba 33 b9
00b0 34 a6 f9 bf bf 99 5f de 76 34 a0 ea be 38 b6 17
00c0 8d e7 c5 5d 5b 22 28 d0 e7 7b b5 35 17 65 22 22
00d0 2d d4 48 39 58 0b 9d 20 b6 12 29 13 12 60 5b de
00e0 f2 99 e0 ed 7c a0 08 16 49 90 95 0e 78 96 47 12
00f0 04 3c 88 1d 74 90 4a dc 54 b7 f9 7b 2c f7 e9 ba
0100 aa ff af 5e 34 b8 ea 45 f4 cf 74 b6 50 d6 7a f4
0110 38 6e 28 57 cf e3 d7 dd 88 93 3b 8c 05 95 84 58
0120 69 5e 94 79 20 39 52 17 c3 a9 00 63 4b ed 44 41
0130 8b 4c 9d 05 d5 c1 1b ca 2b fa 2d 2b c2 3e 6c 12
0140 7e fd d2 9d da 25 51 e4 51 5a 89 85 85 d9 aa e8
0150 d3 ab 02 40 3c 54 65 36 b7 e4 b0

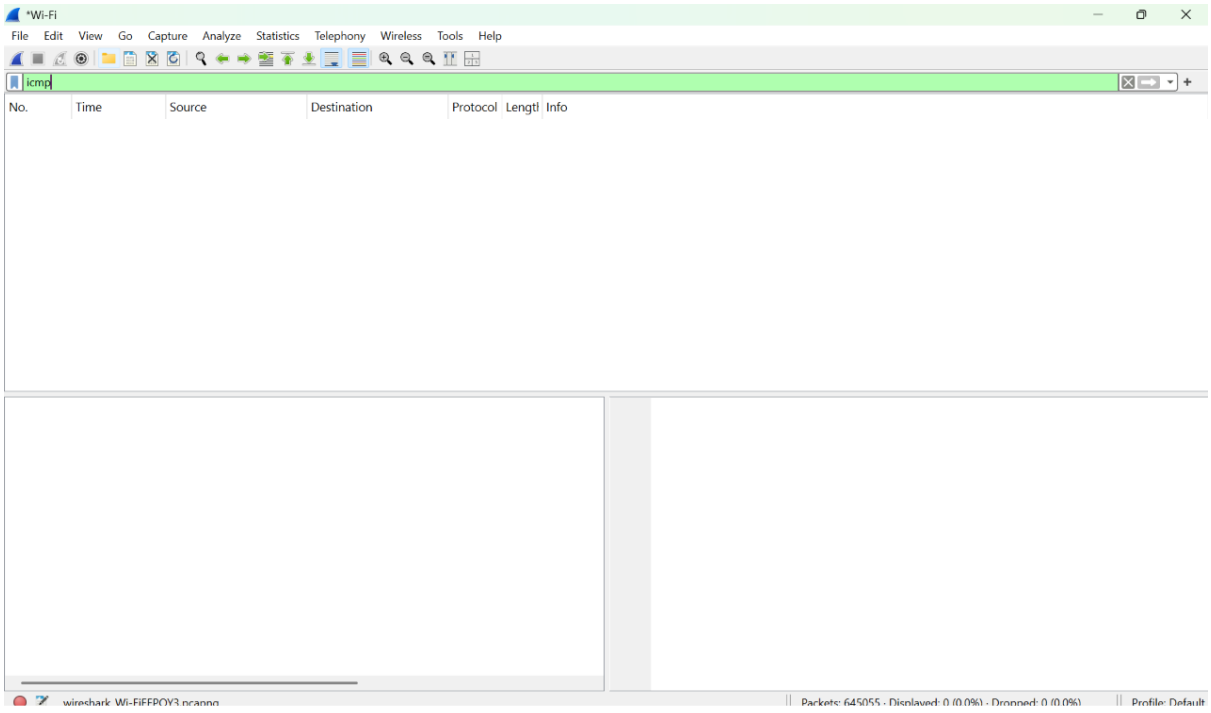

Urgent Pointer: 0
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (293 bytes)
Transport Layer Security
[Stream index: 0]
TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol

UDP

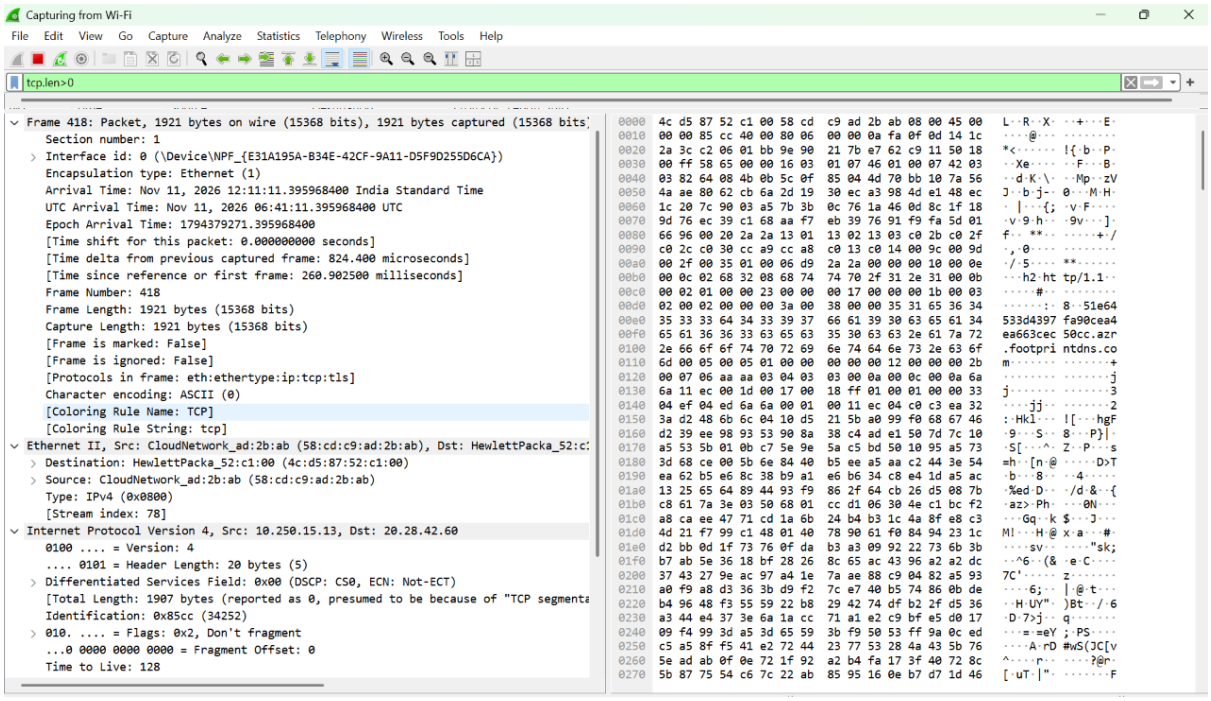

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Frame 1816: Packet, 358 bytes on wire (2864 bits), 358 bytes captured (2864 bits) on i
Section number: 1
Interface id: 0 (\Device\NPF_{E31A195A-B34E-42CF-9A11-D5F9D255D6CA})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 11:56:21.580875800 India Standard Time
UTC Arrival Time: Nov 11, 2026 06:26:21.580875800 UTC
Epoch Arrival Time: 1794378381.580875800
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 481.200 microseconds]
[Time delta from previous displayed frame: 481.200 microseconds]
[Time since reference or first frame: 1.571211500 seconds]
Frame Number: 1816
Frame Length: 358 bytes (2864 bits)
Capture Length: 358 bytes (2864 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:udp:mdns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: Apple_16:c1:d1 (a4:c6:f0:16:c1:d1), Dst: IPv6mcast_fb (33:33:00:00:0
Destination: IPv6mcast_fb (33:33:00:00:00:fb)
.... ..1. .... = LG bit: Locally administered address (this is NO
....1 .... = IG bit: Group address (multicast/broadcast)
Source: Apple_16:c1:d1 (a4:c6:f0:16:c1:d1)
.... ..0. .... = LG bit: Globally unique address (factory default
....0 .... = IG bit: Individual address (unicast)
Type: IPv6 (0x86dd)
[Stream index: 702]
Internet Protocol Version 6, Src: fe80::8bb:b828:ffe4:71e7, Dst: ff02::fb
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)
User Datagram Protocol: Protocol
Packets: 96601 · Displayed: 61322 (63.5%) · Dropped: 0 (0.0%)
Profile: Default

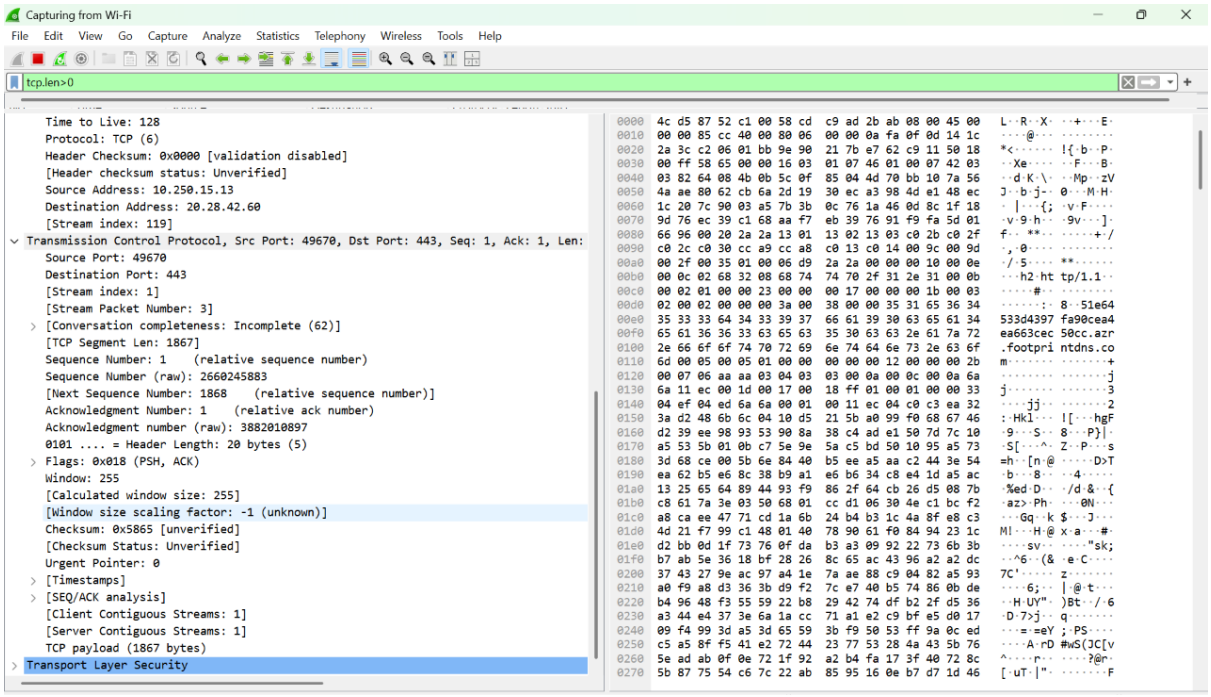

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Type: IPv6 (0x86dd)
[Stream index: 702]
Internet Protocol Version 6, Src: fe80::8bb:b828:ffe4:71e7, Dst: ff02::fb
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Nc
.... 1000 0000 1001 0000 0000 = Flow Label: 0x80900
Payload Length: 304
Next Header: UDP (17)
Hop Limit: 255
Source Address: fe80::8bb:b828:ffe4:71e7
Destination Address: ff02::fb
[Stream index: 264]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353
Destination Port: 5353
Length: 304
Checksum: 0x3e22 [unverified]
[Checksum Status: Unverified]
[Stream index: 459]
[Stream Packet Number: 2]
[Timestamps]
UDP payload (296 bytes)
Multicast Domain Name System (response)
Transaction ID: 0x0000
Flags: 0x8400 Standard query response, No error
Questions: 0
Answer RRs: 3
Authority RRs: 0
Additional RRs: 3
Answers
Additional records
[Request In: 1572]
[Time: 116.577200 milliseconds]

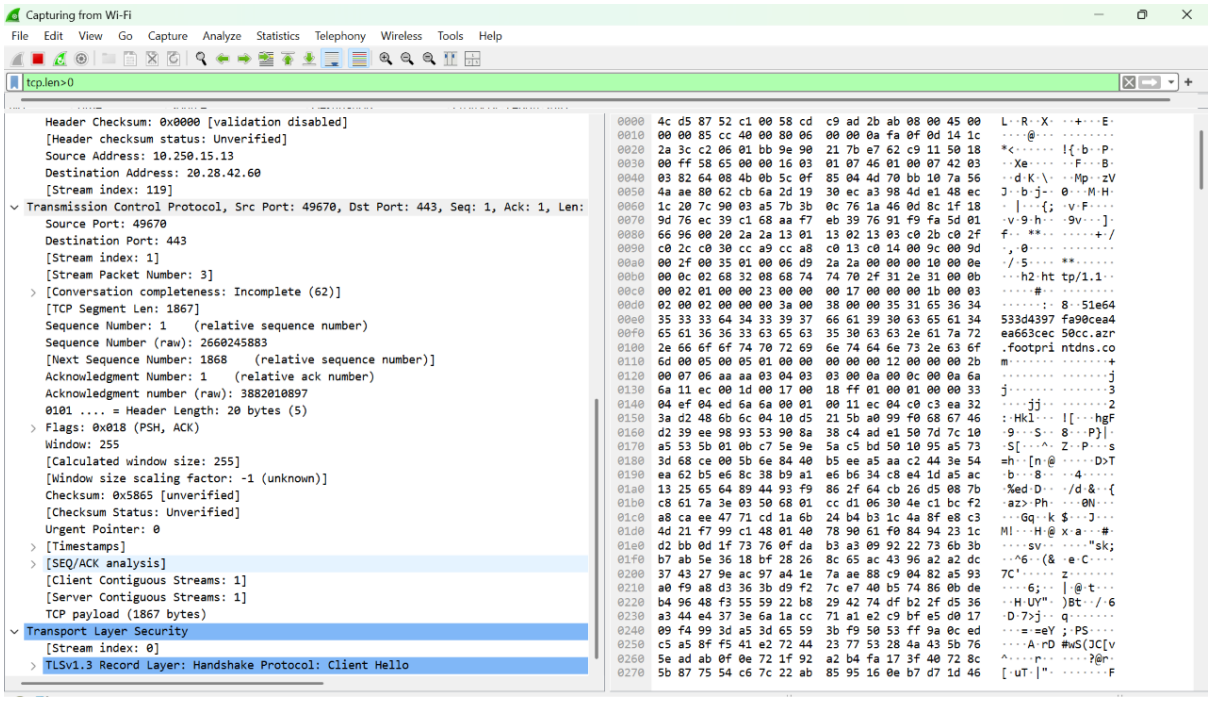
ICMP


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Lengtl Info

TCP.LEN>0


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len>0
Frame 418: Packet, 1921 bytes on wire (15368 bits), 1921 bytes captured (15368 bits)
Section number: 1
Interface id: 0 (\Device\NPF_{E31A195A-B34E-42CF-9A11-D5F9D255D6CA})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 12:11:11.395968400 India Standard Time
UTC Arrival Time: Nov 11, 2026 06:41:11.395968400 UTC
Epoch Arrival Time: 1794379271.395968400
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 824.400 microseconds]
[Time since reference or first frame: 260.902500 milliseconds]
Frame Number: 418
Frame Length: 1921 bytes (15368 bits)
Capture Length: 1921 bytes (15368 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: CloudNetwork_ad:2b:ab (58:cd:c9:ad:2b:ab), Dst: HewlettPacka_52:c1
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Source: CloudNetwork_ad:2b:ab (58:cd:c9:ad:2b:ab)
Type: IPv4 (0x0800)
[Stream index: 78]
Internet Protocol Version 4, Src: 10.250.15.13, Dst: 20.28.42.60
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
[Total Length: 1907 bytes (reported as 0, presumed to be because of "TCP segmenta
Identification: 0x85cc (34252)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128





STEP 6:

TCP PACKET:

Ethernet Header = 14 bytes
IPv4 Header = 20 bytes
TCP Header = 20 bytes
Application Data (TCP Payload) = 1867 bytes
Frame Length = 1921 bytes

Total Header Overhead:

14 + 20 + 20 = 54bytes

Payload Efficiency:

$$\text{Payload Efficiency} = \frac{\text{Application Data}}{\text{Total Frame Length}} \times 100$$

$$= \frac{1867}{1921} \times 100$$

$$= 97.19\%$$

UDP PACKET:

Ethernet Header = 14 bytes
IPv6 Header = 40 bytes
UDP Header = 8 bytes
Application Data (UDP Payload) = 296 bytes
Frame Length = 358 bytes

Total Header Overhead:

$$14 + 40 + 8 = 62 \text{ bytes}$$

Payload Efficiency:

$$\text{Payload Efficiency} = \frac{\text{Application Data}}{\text{Total Frame Length}} \times 100$$

$$= \frac{296}{358} \times 100$$

$$= 82.68\%$$

STEP 7:

Parameter	TCP Packet	UDP Packet
Frame Length	1921 bytes	358 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes (IPv4)	40 bytes (IPv6)
Transport Header Length	20 bytes (TCP)	8 bytes (UDP)
IP Total Length	1907 bytes	344 bytes
Payload Length	1867 bytes	296 bytes